

GIS-Based Analysis of Prospectivity for Kuroko-Type Volcanogenic Massive Sulphide (VMS) Deposits in the Eskay Creek Area, British Columbia, Canada

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ABSTRACT

This study develops a GIS-based prospectivity model for identifying Kuroko-type volcanogenic massive sulfide (VMS) deposits in the Eskay Creek area, British Columbia. By integrating geological, geochemical, and geophysical datasets, the model highlights zones most likely to host these precious metal-rich deposits. Geochemical indicators such as barium (Ba), manganese (Mn), zinc (Zn), silver (Ag), mercury (Hg), and antimony (Sb) were identified as key pathfinders associated with hydrothermal mineralization. Geological controls including andesitic volcanic and volcanoclastic host rocks, along with structural features like dense fault networks, were found to strongly influence mineralization localization. Geophysical data, represented by gravity and magnetic anomalies, further refined prospectivity mapping. The combined weighted overlay approach produced a continuous prospectivity raster that effectively discriminates high-potential areas along the central volcanic belt. Validation against 51 known deposits yielded an area under curve (AUC) of 0.935, confirming strong predictive performance. The model reproduces established geological and geochemical patterns consistent with Eskay Creek-type VMS systems and offers a practical exploration tool for targeting new deposits. Recommendations include focused infill sampling, alteration mineral mapping, and 3D modeling to enhance prediction accuracy. This integrated GIS methodology sets a foundation for regional exploration and can be adapted to similar terranes worldwide.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
GIS	Geographic Information System
VMS	Volcanogenic Massive Sulphide
IDW	Inverse Distance Weighting
ROC	Receiver Operating Characteristic
Ag	Silver
Ba	Barium
Hg	Mercury
Pb	Lead
Sb	Antimony
Zn	Zinc
QGIS	Quantum GIS software

CHAPTER 1: INTRODUCTION

1.1 Background and Importance of the Study

Mineral resources are foundational to the economic development and industrial advancement of many nations, particularly those reliant on primary industries. The ongoing demand for base and precious metals such as copper, lead, zinc, and gold continues to drive exploration in historically productive mining regions. However, these mature districts face a significant challenge: while they are rich in geoscientific data, the discovery of new deposits is increasingly difficult due to resource depletion and the overprinting of geological signals by prior mining activity (Franklin et al., 2005).

Kuroko-type volcanogenic massive sulphide (VMS) deposits are among the most significant exploration targets in felsic-dominated, extensional submarine volcanic arc environments. These deposits are characterized by stratabound accumulations of polymetallic sulphides, commonly pyrite, chalcopyrite and sphalerite and are closely associated with felsic volcanic rocks, hydrothermal alteration zones, and extensional fault structures. For this study, the alteration signature was captured indirectly through geochemical anomaly layers, and structural features were inferred from geophysical data due to the absence of explicit fault maps (Franklin et al., 2005). The Eskay Creek area in northwestern British Columbia, located within the Stikine Terrane, is a compelling example of such a deposit, formed in a classic submarine volcanic arc setting.

It hosts the world-renowned Eskay Creek deposit, one of the highest grade Au and Ag VMS deposits ever discovered (Figure 1.1). This deposit formed during the Middle Jurassic in a tectonically active back-arc or intra-arc basin (Barrett & Sherlock, 1996; Sherlock et al., 1999). Despite Eskay Creek's exceptional mineral endowment, further

discoveries in the region have remained limited. This highlights two key challenges in mineral exploration that directly inform the motivation for this study: The lack of integration between geological, geochemical, and geophysical datasets has limited the ability to understand the spatial relationships that control mineralization. This fragmentation reduces the precision and predictive power of mineral prospectivity mapping, even when GIS is employed.



Figure 1.1: Location map of Eskay Creek in northwestern British Columbia, showing the study area. Adapted from Tornos et al., 2019).

Secondly, existing studies in the Eskay Creek area have not fully integrated these datasets into a unified GIS-based predictive model using the specific combination of methods presented in this study, which are based on a conceptual model tailored to the Eskay Creek deposit. As a result, prospectivity analyses in this region may overlook important correlations and subtle indicators of concealed mineralization.

To address these challenges, this research applies a data-driven approach that integrates spatial analysis and statistical processing. The primary spatial analysis was conducted using QGIS, which was used to prepare, reclassify, and combine the evidential data layers through a weighted overlay approach. Statistical parameters, including mean, standard deviation, and percentiles, were calculated for geochemical data within QGIS to define thresholds for anomaly mapping. The model's predictive performance was then quantitatively assessed using validation techniques recommended in the literature, which provided a statistically rigorous evaluation of the results

1.2 Brief Literature Review

1.2.1 Kuroko-Type VMS Deposits

Kuroko-type VMS deposits typically form in submarine volcanic arc settings dominated by felsic volcanic rocks. These deposits develop in extensional tectonic environments where hydrothermal fluids migrate through permeable pathways, precipitating stratabound accumulations of polymetallic sulphides near the seafloor. The distribution of favorable host rocks, especially rhyolitic lithologies, and the presence of alteration zones rich in minerals such as silica, sericite, and chlorite, play crucial roles in localizing mineralization (Franklin et al., 2005; Simmons et al., 2005). The ore zones commonly exhibit vertical and lateral zoning,

with dominant sulphide minerals including sphalerite, chalcopyrite, and galena, reflecting the complex hydrothermal processes involved (Singer, 1995; Franklin et al., 2005).

1.2.2 The Eskay Creek Area

The Eskay Creek deposit in northwestern British Columbia is one of the most significant precious metal-rich VMS systems in Canada. Hosted in the Hazelton Group of the Stikine Terrane, it is notable for its bimodal volcanic setting and association with sedimentary rocks of the Salmon River Formation. The deposit is characterized by high-grade gold and silver mineralization, making it both economically and scientifically valuable. Its complex stratigraphy, hydrothermal alteration, and tectonic context interpreted as part of the Eskay Rift are illustrated in Figure 1.2 and highlight the deposit's exceptional geological setting (Alldrick et al., 1996; Barrett & Sherlock, 1996)

Fluid inclusion and stable isotope data suggest a volcanogenic origin influenced by evolved hydrothermal fluids (Sherlock et al., 1999).

1.2.3 GIS in Mineral Prospectivity

GIS has become integral to modern mineral exploration, providing the essential platform for integrating large, heterogeneous datasets. The application of GIS for mineral prospectivity mapping was fundamentally advanced by the work of Carranza (2009), who provided a systematic framework for analyzing and integrating geological, geochemical, and geophysical data layers. His methods, building on foundational concepts like those of Bonham-Carter (1994), leverage techniques such as multi-criteria decision analysis (MCDA) and fuzzy logic to identify zones of high mineral potential. A key contribution of this body of work is its focus on quantifying and managing the inherent uncertainty in spatial models. This emphasis has directly informed recent research, which continues to develop more robust

validation techniques and ways to account for workflow-induced uncertainty in prospectivity models (Zuo et al., 2021).

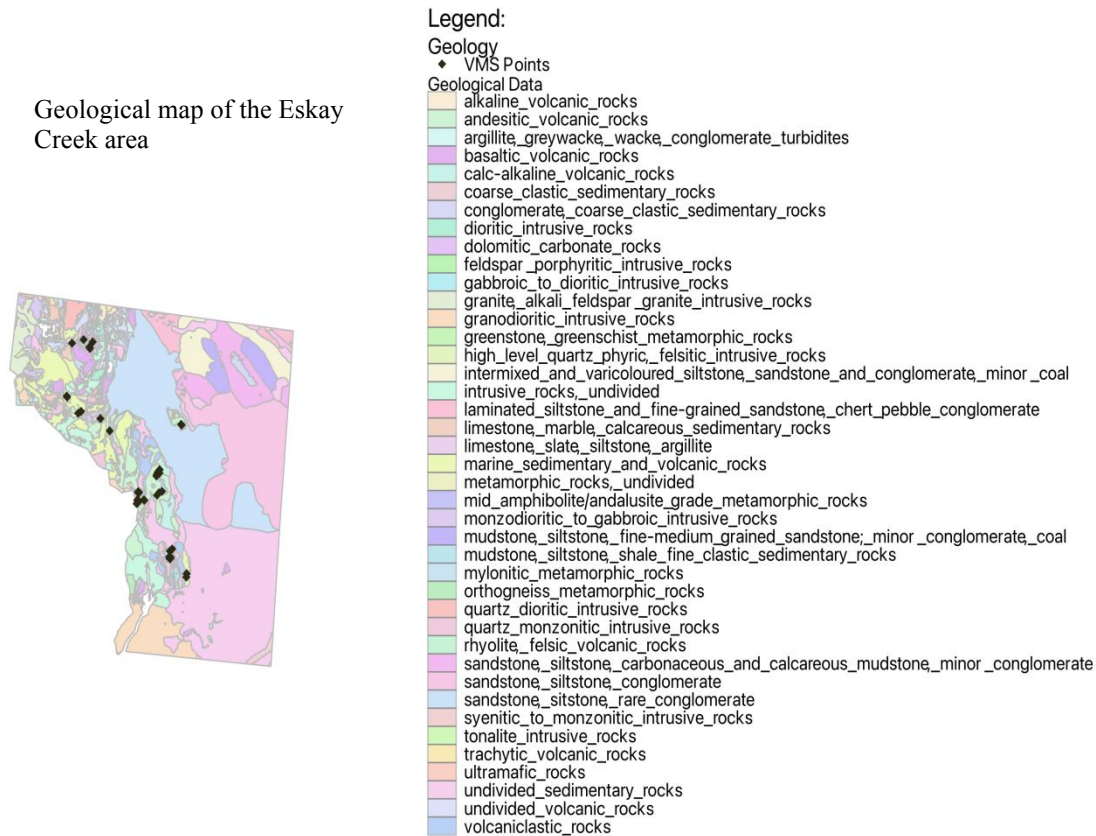


Figure 1.2: *Stratigraphy and geology of the Eskay Creek deposit.*

1.2.4 Integration Limitations

Despite the capabilities of GIS, few studies in the Eskay Creek region systematically integrate geological, geophysical, and geochemical data within a single framework. Many analyses remain isolated, limiting the full potential of spatial models. The lack of structured, multidisciplinary workflows hampers effective target generation, especially in geologically complex terrains like those of the Stikine Terrane. A more unified GIS-driven approach could improve mineral prospectivity assessments in this underexplored region (Zuo et al., 2021).

This study aims to address this gap by integrating geological, geophysical, and geochemical datasets into a structured GIS-based prospectivity model for the Eskay Creek region

1.3 Research Gap and Aim

1.3.1 Research Gap

Despite the availability of high resolution geological and geophysical datasets from agencies such as the British Columbia Geological Survey (BCGS), these were often underutilized in combination. Existing studies typically focused on single data domains either structural geology, geochemistry, or magnetics rather than applying an integrated approach tailored to Kuroko-type VMS systems.

1.3.2 Aim

This study aimed to develop a GIS-based prospectivity model to delineate areas favourable for Kuroko-type VMS deposits in the Eskay Creek area by integrating geological, geochemical, and geophysical datasets using spatial and statistical analysis tools.

1.4 Significance of the Study

This research presents a practical GIS-based framework for exploring volcanogenic massive sulphide (VMS) deposits in mature and geologically complex districts. By using QGIS for spatial analysis and the Python console for data processing and automation, the study provides a replicable methodology that helps reduce exploration uncertainty and improves the identification of prospective VMS targets.

The framework integrates geochemical, geophysical, and geological datasets in a systematic and reproducible manner, offering a clear approach for delineating areas with high mineral potential. In addition to its practical application, this study contributes to the scientific

understanding of VMS exploration, providing a workflow that can guide future research and support applied mineral exploration in well-studied terrains.

Through the integration of geological, geochemical, and geophysical data in QGIS, and the application of spatial analysis tools available through its core and plugin libraries (e.g, raster reclassification, weighted overlay, interpolation, and density mapping via SAGA and GDAL), this research enhances mineral prospectivity modelling and addresses key challenges associated with deposit discovery in well-explored terrains.

1.5 THESIS STRUCTURE

This thesis is organized as follows:

Chapter 2 provides a review of literature relevant to Kuroko-type VMS deposits, the Eskay Creek geological context, and GIS applications in mineral exploration.

Chapter 3 describes the datasets and outlines the GIS-based methodology used in spatial and statistical modelling.

Chapter 4 presents the results of data integration and prospectivity mapping.

Chapter 5 discusses the outcomes in relation to geological models and exploration implications.

Chapter 6 concludes the study and suggests directions for future research and practical application.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Kuroko-Type VMS Deposits

Kuroko-type volcanogenic massive sulfide (VMS) deposits are a variety of VMS deposit named after the famous Miocene-aged deposits of the Kuroko mining district in Japan. These deposits consist of stratiform accumulations of sulfide minerals formed from hot, metal-rich hydrothermal fluids discharged onto or near the seafloor. Early research established the key features of Kuroko deposits, with foundational descriptions of the characteristic "black ore" mineralogy (Kinoshita, 1943, as cited in Collins, 1950). These polymetallic deposits are known for their economically significant concentrations of copper, zinc, lead, gold, and silver. Western researchers later built upon this knowledge to create the comprehensive descriptive models used today (Ohmoto & Skinner, 1983; Singer, 1986).

2.1.1 Key Geological Features

The classic model for Kuroko-type deposits provides the foundation for understanding their specific geological characteristics, as illustrated in Figure 2.1, which is based on the influential USGS Model 28a. This model emphasizes the association with marine felsic to intermediate volcanic rocks in island arc successions that exhibit bimodal volcanism (Yamada & Yoshida, 2011). These deposits form in extensional tectonic settings, such as back-arc basins, where seafloor spreading facilitates the movement of hydrothermal fluids (Ohmoto & Skinner, 1983). A shallow sub-volcanic magmatic intrusion provides the heat engine for the hydrothermal fluid convection system that drives mineralization (Ohmoto & Skinner, 1983).

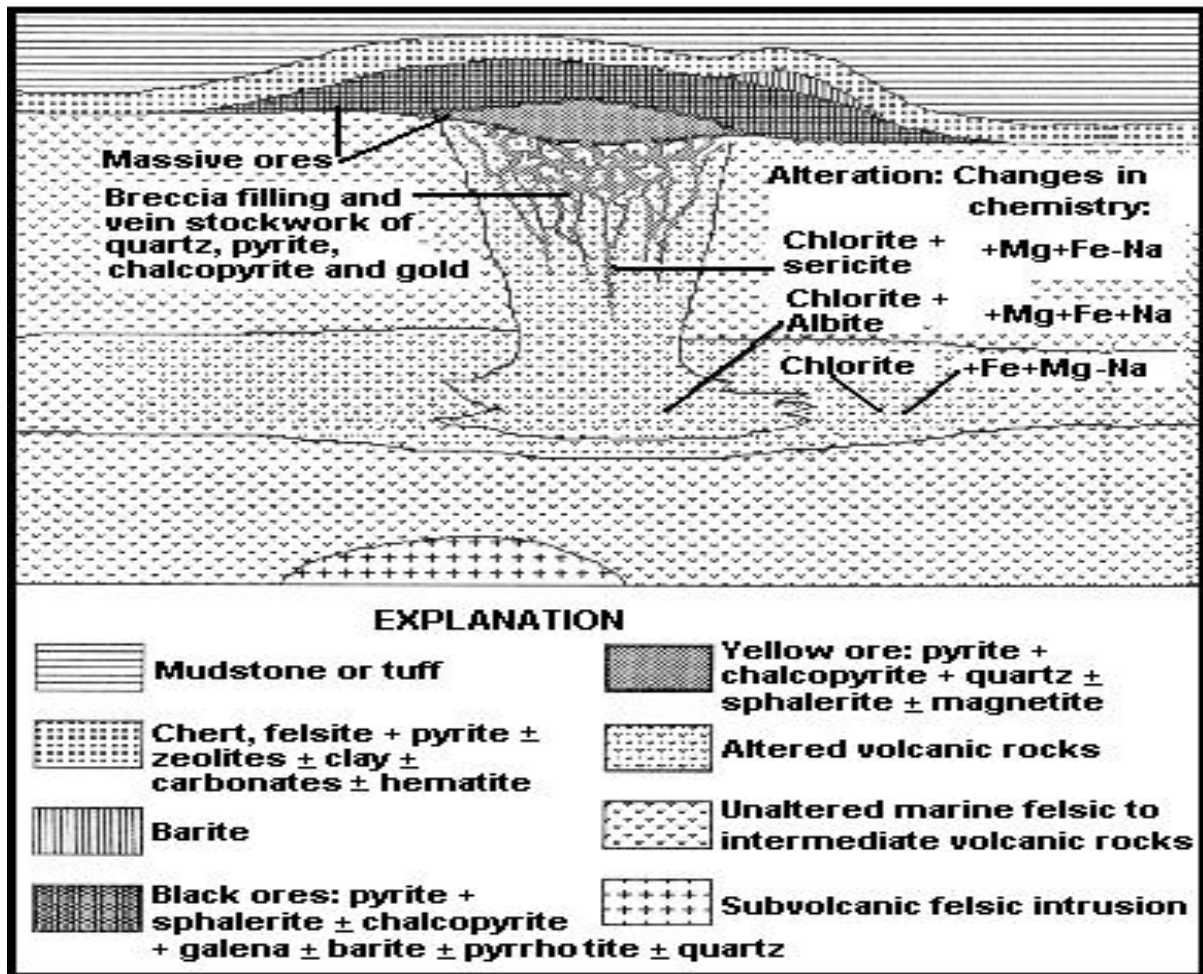


Figure 2.1: *Classic model of a Kuroko-type massive sulfide deposit (Adapted from USGS Model 28a, Singer, 1986). The diagram illustrates the association with felsic volcanics, the characteristic ore zonation (including copper-rich stringer ore, yellow ore, and black ore), the overlying mudstone caprock, and surrounding alteration halos.*

As shown in Figure 2.1, Kuroko deposits are renowned for their distinct metal zoning, which is a product of fractional precipitation as the hydrothermal fluids cool upon mixing with seawater (Hannington, 2014). This results in a copper-rich stockwork feeder zone, or yellow ore, forming beneath the main deposit. This zone then grades laterally and upward into the polymetallic, fine-grained black ore (kuroko), which is composed of sphalerite, galena, and barite. The stratabound, mound or lens-shaped bodies of black ore were formed from seafloor hydrothermal venting (Mindat.org, 2025; USGS, 1995).

Host volcanic rocks typically include rhyolites, dacites, tuffs, and subordinate basalts, which are interbedded with volcanic sediments, such as organic-rich mudstones. These impermeable mudstone layers often act as a cap rock, trapping mineralizing fluids and helping to preserve the deposit, a feature evident in the model (Ohmoto & Skinner, 1983).

Alteration patterns in Kuroko deposits also exemplify the model, with concentric halos clearly defined. A footwall silicic alteration core develops around the fluid conduit, enveloped by sericite- and chlorite-rich zones (Singer, 1986). This concentric pattern is a direct result of the thermal and chemical gradients of the hydrothermal system and serves as a vital exploration tool.

2.1.2 Metal Content and Economic Importance

These deposits have significant amounts of copper, zinc, and lead, along with valuable amounts of gold and silver. The high gold and silver content makes Kuroko-type deposits especially important compared to other VMS types. For example, the Eskay Creek deposit in British Columbia is known for very high gold and silver grades, making these deposits good targets for exploration (Sherlock et al., 1999; Childe, 1996).

2.1.3 Typical Settings

Kuroko-type deposits form in submarine environments that are tectonically active, typically within island arc and back-arc extensional settings. These deposits are commonly hosted in felsic to intermediate volcanic and volcanoclastic rocks, often associated with submarine rhyolitic domes and marine sedimentary rocks, including organic-rich mudstones. Active

volcanism and faulting in these environments create essential pathways for the circulation of metal-rich hydrothermal fluids, which deposit sulfides at or near the seafloor (Alldrick et al., 1996; Barrett & Sherlock, 1996).

Beyond describing their formation, how do these predictable characteristics make Kuroko-type deposits so suitable for GIS-based exploration modeling? The clear patterns of alteration and mineralization, together with specific host rock types and chemical signatures, make Kuroko-type deposits suitable for GIS-based exploration modelling. They have predictable relationships between geology, alteration, and metal content that help create maps showing where deposits might be. GIS allows different types of data like geology, chemistry, and geophysics to be combined to find targets more efficiently (Bonham-Carter, 1994; Zuo et al., 2021).

2.2 Geological Setting of the Eskay Creek Area

The Eskay Creek deposit is located in the Stikine Terrane of northwestern British Columbia. This terrane is an important tectonostratigraphic unit in the Canadian Cordillera formed by the joining of island arc complexes during the Mesozoic era (Alldrick, Nelson, & Barresi, 1996). The region mainly consists of the Hazelton Group, which is a Middle to Upper Jurassic sequence of volcanic and sedimentary rocks that hosts many metal deposits (Barrett & Sherlock, 1996).

The deposit lies within the Eskay Rift, a local area of extension formed during the Early to Middle Jurassic. This created the right volcanic and structural conditions for VMS deposits to form (Alldrick et al., 1996; Brueckner et al., 2021). The rock layers include parts of the Lower and Upper Hazelton Groups, but most mineralization happens in the Contact Unit.

This unit has rhyolitic volcanic rocks mixed with carbon-rich mudstones (Mortensen et al., 2003). These rocks provided the volcanic material and reducing environment needed to form and preserve massive sulfide mineralization.

Eskay Creek is a precious metal-rich VMS system. It has both massive and semi-massive sulfide mineral zones that are bedded or replace the surrounding rock. The main sulfide minerals are pyrite, sphalerite, and chalcopyrite. Important gold and silver minerals include electrum, argentite, and native silver (Sherlock, Roth, Spooner, & Bray, 1999). This shows how Eskay Creek is special because of its high gold and silver compared to other VMS deposits.

Dating shows that mineralization occurred about 175 million years ago during rifting and volcanism (Mortensen et al., 2003). Although the area was explored off and on in the 20th century, the deposit was only found in the 1980s. Mining started in 1994 and continued until 2008. During this time, about 3.3 million ounces of gold and 160 million ounces of silver were produced, making Eskay Creek one of the richest gold-silver VMS deposits in the world (Sherlock et al., 1999).

2.3 GIS Applications in Mineral Prospectivity Mapping

The combination of rift tectonics, suitable host rocks, and well-preserved mineralization makes Eskay Creek a classic example for studying mineral exploration and GIS prospectivity modelling. GIS-based mineral prospectivity mapping has become an important tool in modern mineral exploration. It allows different geological data such as rock types, chemistry, geophysics, and structures to be combined and analyzed to find areas with higher chances of

mineral deposits. Common GIS methods include Multi-Criteria Decision Analysis, fuzzy logic, and Weight of Evidence (Bonham-Carter, 1994).

Additional methods such as logistic regression, Bayesian network modelling, and data-driven machine-learning classifiers (e.g., random forest, support-vector machine) have been increasingly applied to mineral prospectivity mapping to handle nonlinear relationships between variables (Porwal et al., 2003; Carranza, 2009; Harris, 2017). These techniques allow the integration of large spatial datasets and can complement knowledge-driven approaches in early-stage exploration.

GIS helps show where patterns related to mineral deposits occur and supports making data-driven decisions. Data like rock boundaries, alteration zones, faults, and chemical anomalies can be weighted and analyzed to estimate where minerals might be. In conceptual-model-driven mapping, these spatial layers represent geological processes such as fluid pathways, permeability contrasts, and geochemical dispersion patterns. Carranza (2011) emphasizes that prospectivity analysis should begin with a conceptual geological model that defines the controlling factors of the deposit type and then translates them into measurable spatial criteria.

However, GIS methods also have challenges, mainly related to how reliable the results are. Zuo et al. (2021) reviewed the main uncertainties affecting GIS prospectivity mapping. These include data uncertainty, model uncertainty, interpretation uncertainty, and integration uncertainty. Data uncertainty comes from errors in maps, differences in data resolution, and mistakes in sampling or location. Model uncertainty comes from subjective choices in selecting data, assigning weights, or setting thresholds in the GIS. Interpretation uncertainty relates to how the maps are understood and used, which can vary by user knowledge. Integration uncertainty happens when combining different models and data, which can spread errors.

Carranza (2009) and Harris (2017) highlight that validation techniques such as cross-validation, receiver-operating-characteristic (ROC) curves, and comparison with known deposit locations are critical for assessing map performance. Sensitivity analysis can determine how individual criteria affect the final output, reducing model bias and increasing reliability.

These uncertainties can affect the trustworthiness of prospectivity maps. If not handled, they may miss good targets or suggest poor ones, affecting exploration decisions. To reduce these risks, Zuo et al. (2021) suggest methods such as sensitivity testing, checking results with other data, and using multiple models together to get more reliable results. Also, clear explanation of how models are built and what assumptions are made can increase confidence. Transparent documentation of data sources, weighting schemes, and criteria selection is recommended to ensure reproducibility and reliability.

Overall, GIS prospectivity mapping is a powerful tool, but its limits should be known and managed. Doing so improves the quality of exploration plans and lowers the chance of costly mistakes. Using conceptual models based on specific deposit types, like Kuroko-type VMS, helps make prospectivity maps more realistic and useful.

2.4 Conceptual Models for VMS Prospectivity

Conceptual models are important tools in mineral exploration (Figure 2.2). They provide a clear geological framework that helps choose which features to use in prospectivity mapping. For VMS systems like Kuroko-type deposits such as Eskay Creek, these models include the main geological, chemical, and structural factors that control mineralization. Conceptual models also incorporate temporal and spatial relationships between volcanic events,

hydrothermal alteration, and sulfide deposition, which are critical for defining prospective zones in GIS workflows (Carranza, 2009; Brueckner et al., 2021).

Kuroko-type VMS deposits usually form in underwater volcanic settings. They are often linked with calc-alkaline felsic volcanism in arc extension environments. These deposits have distinct zones of hydrothermal alteration, clear metal zoning, and sulfide bodies that are layered or stockwork and rich in base and precious metals. Eskay Creek is hosted in the Hazelton Group, a Lower Jurassic volcanic and sedimentary sequence that experienced rifting in the Eskay Rift (Alldrick et al., 1996). The mineralization shows typical Kuroko features including rhyolite volcanic rocks, strong hydrothermal alteration, and metal zoning with gold and silver-rich barite and sulfides (Barrett & Sherlock, 1996; Sherlock et al., 1999). Alteration zoning at Eskay Creek typically progresses from an inner silica-sericite-pyrite halo surrounding massive sulfides to outer chlorite and carbonate zones. These zoned alteration halos can be converted into GIS raster layers or buffer layers for prospectivity mapping, providing a quantitative representation of hydrothermal influence (Childe, 1996).

In GIS prospectivity mapping, conceptual models guide which spatial data are important for finding VMS deposits. For example, felsic volcanic rocks hosting Eskay Creek are shown as lithology layers in GIS. Faults and volcanic-related structures act as fluid paths and can be mapped using distance or density layers (Brueckner et al., 2021). Hydrothermal alteration can be mapped with chemical indexes like the Ishikawa Alteration Index using portable XRF or ICP-MS data (Barrett & Sherlock, 1996). Structural complexity, such as fault intersections, shear zones, and synvolcanic fractures, can be weighted according to proximity or density to reflect hydrothermal fluid focusing. These GIS layers integrate directly with knowledge-driven multi-criteria decision models for VMS targeting (Carranza, 2011).

Conceptual models help select and combine spatial features that make geological sense in the maps. Important features for VMS include proximity to felsic volcanic centers, stratigraphic position in rift volcanic sequences, presence of faults, chemical anomalies in metals and alteration indicators, and geophysical signals such as conductivity highs or magnetic lows. By matching GIS layers to known geological controls, conceptual models improve how well data-driven methods like Weight-of-Evidence, logistic regression, or fuzzy logic work. Combining conceptual knowledge with empirical data also reduces interpretation and integration uncertainties (Zuo et al., 2021). This helps ensure that statistical patterns reflect real geology.

In practice, Carranza (2009) emphasizes transforming these conceptual geological controls into spatially explicit criteria, assigning probabilities or suitability scores to each GIS layer based on observed correlations with known deposit occurrences, thereby creating a statistically rigorous foundation for prospectivity mapping.

The Eskay Creek example shows how a clear conceptual model based on stratigraphy, structure, and age data (Childe, 1996) can improve exploration results. For example, the deposit is located at the contact between rhyolite and mudstone, a key spatial marker that can be used to find similar deposits. Additional GIS layers can include proximity to alteration halos, host-rock facies distribution, hydrothermal vent pathways, and known sulfide occurrences. These layers are derived directly from geological maps, geochemistry, and geophysics, and collectively represent the spatial manifestation of the conceptual model (Carranza, 2011).

In summary, conceptual models form the foundation of VMS exploration and connect geological understanding with GIS tools. For Kuroko-type systems like Eskay Creek, they help focus on features that control mineralization, improve the use of spatial data, and raise

the quality and confidence of prospectivity maps. Applying this framework ensures that prospectivity outputs are geologically meaningful and directly tied to known deposit-forming processes, improving both target ranking and exploration efficiency in areas with similar arc-rift volcanic sequences. Following the structured approach of Carranza (2011), conceptual models are first translated into spatial layers representing key geological, chemical, and structural controls. These layers are then integrated using a knowledge-driven multi-criteria evaluation, validated against known deposits, and finally used to identify high-potential exploration targets. This systematic framework ensures that each step, from conceptual understanding to GIS-based mapping is geologically justified and reproducible.

Figure 2.2 provides the conceptual workflow for GIS-based VMS prospectivity mapping. In the following chapter, this framework is operationalized through a multi-source, QGIS-based methodology that integrates geological, geochemical, and geophysical datasets to predict VMS prospectivity in the Eskay Creek area.

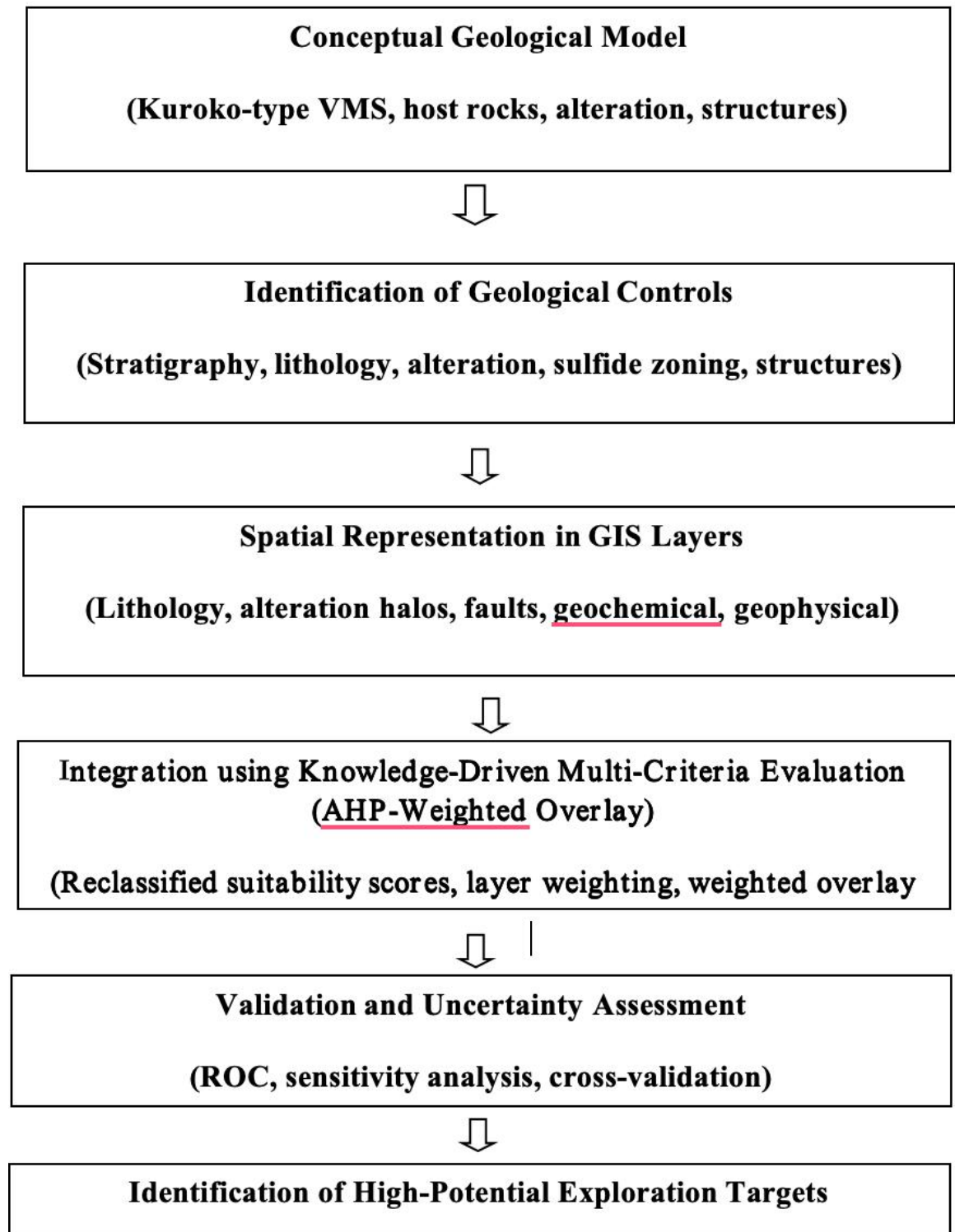


Figure 2.2: *Conceptual Workflow for GIS-Based VMS Prospectivity Mapping*

CHAPTER 3: DATA AND METHODOLOGY

This chapter provides a detailed explanation of the datasets used in the study and the methods that were followed to conduct a GIS-based prospectivity analysis for Kuroko-type volcanogenic massive sulphide (VMS) deposits in the Eskay Creek area. The analysis was conducted using a weighted overlay approach in the QGIS software.

3.1 Overview of Datasets

All datasets used in this study were provided by the research supervisor, originating from the British Columbia Geological Survey (BCGS). These datasets include: mineral deposit occurrence locations, geological maps, a multi-element geochemical dataset from stream sediments, and geophysical datasets (airborne gravity residual, airborne magnetic, and magnetic data). These datasets are all specific to the Eskay Creek area and its surrounding region. They serve as the foundation for the spatial analysis and GIS-based prospectivity mapping in the Eskay Creek area of British Columbia, Canada.

3.1.1 Geological Datasets

The geological dataset for the Eskay Creek area was provided as a vector polygon shapefile, containing detailed lithological and stratigraphic information. This dataset included attributes such as stratigraphic group and formation names, rock types, ages, and tectonic province classifications. To facilitate meaningful spatial analysis and integration with other datasets, the raw lithological units were reclassified into simplified, standardized categories. These categories included: felsic, mafic, volcanic sedimentary and other. This reclassification reduced complexity, improved interpretability, and allowed for the assignment of suitability

scores based on the known affinities of these rock types for hosting VMS mineralization (Barrett & Sherlock, 1996; Childe, 1996). The geological polygons, already clipped to the study area, were then used as input layers in GIS spatial analyses (Bonham-Carter, 1994). Lithological categories were assigned weights in a suitability model, reflecting their relative potential for VMS deposit occurrence. This enabled the creation of a lithological suitability raster for integration into the overall prospectivity analysis. Structural data such as faults and folds were not explicitly available in the geological dataset. Therefore, structural controls were inferred indirectly through geophysical data, as described in the geophysical methodology section (Alldrick et al., 1996).

3.1.2 Geochemical Datasets Processing and Normalization

The stream sediment geochemical dataset was preprocessed and integrated following a systematic spatial workflow to ensure compatibility with other evidence layers used in the mineral prospectivity model. The raw data, derived from the British Columbia Geological Survey, included concentrations of multiple elements sampled across the study area. All geochemical point data were projected to EPSG:26909 (NAD83 / UTM Zone 9N) to maintain spatial consistency with other datasets.

Prior to analysis, the dataset was visually inspected and statistically reviewed to identify and remove potential outliers. Descriptive statistics, including mean, standard deviation, and coefficient of variation, were computed for each element. These were used to assess data distribution and element variability, following the approach recommended by Carranza (2009).

To generate continuous geochemical surfaces, Inverse Distance Weighting (IDW) interpolation was applied in QGIS. IDW was selected after evaluating Kriging and Spline methods; it provided the most geologically consistent representation of local anomalies while minimizing oversmoothing artifacts. The resulting interpolated rasters represent continuous spatial distributions of element concentrations.

Anomaly Standardization (0- 1 Scale)

The interpolated geochemical rasters for each element were standardized to a continuous 0-1 scale using min-max normalization. In this procedure, the lowest raster value was assigned a score of 0 and the highest a score of 1, with intermediate values scaled proportionally. This ensured all geochemical layers were expressed in comparable, dimensionless units prior to integration with structural, lithological, and geophysical datasets in the weighted overlay analysis. The normalization step also reduced bias arising from differences in elemental concentration ranges.

Weighted Integration of Key Elements

Following normalization, selected elements known to be geochemically indicative of mineralization in the Eskay Creek area specifically Mn, Ba, Zn, Hg, and Ni were integrated using weighted overlay analysis. Each element's normalized raster was multiplied by its corresponding weight (as defined in Table 4.2), reflecting its relative significance in the mineralization model. The weighted rasters were then summed to generate a composite geochemical favorability map, which was later incorporated into the final prospectivity model.

Interpretation of Geochemical Anomalies

Areas of elevated composite geochemical scores correspond to zones with anomalous element enrichment, which are interpreted as potential hydrothermal alteration or mineralization zones. These geochemical patterns spatially coincide with the known mineralized trends of the Eskay Creek Formation and thus provided critical input for subsequent integration with geological and structural datasets.

3.1.3 Geophysical Datasets and Structural Inference

Geophysical raster datasets, including total magnetic intensity (area_mag), residual gravity (area_grres), and the first vertical derivative of magnetic data (area_fd), were integrated into the GIS-based prospectivity analysis for the Eskay Creek area. These datasets cover the Eskay Creek area and reflect subsurface magnetic and gravitational features correlated with known mineralization, such as the low magnetic signatures associated with alteration zones. The first vertical derivative raster was examined to identify sharp magnetic gradients indicative of structural lineaments, which correspond to faults or lithological contacts. The structural lineaments digitized in this study correspond to documented fault zones near the Eskay Creek deposit, which are believed to have acted as key fluid pathways. Residual gravity and total magnetic intensity rasters were analyzed to detect variations in subsurface density and magnetic properties, which are associated with different lithologies and alteration zones important for VMS mineralization (Brueckner et al., 2021; Sherlock et al., 1999).

Due to the absence of explicit structural data like mapped faults, potential structural controls were inferred indirectly. Linear features visible in the area.fd raster were manually digitized

as vector line shapefiles using the Add Line Feature tool in QGIS. These digitized lineaments represent inferred faults and contacts. Buffer zones ranging from 100 to 500 meters were created around the digitized lineaments to generate proximity rasters. These rasters assign higher suitability scores to areas closer to the inferred structural features, reflecting the geological understanding that proximity to faults often controls mineralization (Carranza, 2011).

3.2 Data Preprocessing and Integration

All datasets were projected to EPSG:26909 (NAD83 / UTM Zone 9N) to ensure spatial alignment and compatibility between layers (Carranza, 2008). Each dataset was clipped to the Eskay Creek study area to maintain a consistent spatial extent for analysis.

The geological dataset was simplified by reclassifying lithological units into standardized categories (felsic, mafic, volcanic sedimentary, and other), reducing complexity and enhancing interpretability.

For the geochemical data, point samples were used to generate continuous surface maps through Inverse Distance Weighting (IDW) interpolation. This process was conducted directly in QGIS using the built-in Interpolation tool, which estimates elemental concentration values across unsampled areas based on the spatial proximity of known sample points. The resulting interpolated rasters were used to represent spatial variation in geochemical anomalies relevant to VMS mineralization.

All raster layers were resampled to a uniform spatial resolution of 100×100 m, balancing spatial detail with computational efficiency. This ensured that all datasets were compatible for multi-layer raster operations such as reclassification and weighted overlay.

Finally, the datasets were integrated using spatial joins and overlays within QGIS, forming the composite input for the GIS-based prospectivity analysis. This preprocessing workflow ensured spatial consistency, data comparability, and analytical reproducibility throughout the subsequent stages of analysis (Carranza, 2009; Bonham-Carter, 1994).

3.3 Spatial Analysis Techniques and Suitability Scoring

This research employed various GIS spatial analysis methods to examine the spatial relationships of the geological, geochemical, and geophysical data with the known mineral occurrences (Bonham-Carter, 1994). The techniques included: buffer analysis around known VMS points to assess proximal relationships; kernel density estimation to visualize concentrations of geochemical anomalies and inferred structural features. Raster reclassification of geophysical data was performed to standardize suitability scores for prospectivity. Each geophysical raster dataset was analyzed to identify value ranges associated with mineralization-related features

These raw raster values were then reclassified into a normalized suitability scale from 0 to 1, where: 0 represents areas least favorable for hosting VMS deposits, and 1 represents areas most favorable (Bonham-Carter, 1994; Zuo et al., 2021). This was done using GIS tools in QGIS such as the Raster Calculator or Reclassify tool by assigning input value ranges to corresponding suitability scores on the 0-1 scale. The reclassified rasters could then be combined consistently with other standardized layers during the weighted overlay prospectivity analysis; overlay analysis combining multiple evidence layers to identify areas of potential mineralization. These analyses aimed to detect spatial patterns indicative of favorable conditions for Kuroko-type VMS deposits.

3.4 Prospectivity Mapping Methodology

Note that a separate alteration raster was not created in this study. Instead, the hydrothermal alteration signature was captured indirectly through the geochemical anomaly layers, as elevated concentrations of pathfinder elements like Cu, Zn, Pb, and S are spatially associated with the silica-sericite-chlorite alteration halos typical of Eskay Creek VMS systems (Carranza, 2009). The reclassification of these anomaly rasters therefore served to represent the alteration signal within the model.

3.4.1 Preparing Evidence Layers

The exploration evidence layers were prepared to create standardized, spatially aligned inputs for the weighted overlay analysis. Each dataset was processed as follows:

Geological Layers:

The geological dataset was simplified and categorized into broader lithological groups, reflecting the regional and local controls on VMS mineralization. Lithology classes were reclassified into standardized categories to facilitate integration with other evidence layers (Carranza, 2011). This ensured that the lithology raster could be directly incorporated into the weighted overlay model.

Geochemical Layers:

Geochemical point data for selected pathfinder elements (Mn, Ba, Zn, Hg, Ni) were interpolated to create continuous raster surfaces using the Inverse Distance Weighting (IDW) method in QGIS. The IDW interpolation preserved local geochemical variability while minimizing artifacts compared to Kriging and Spline interpolation. Each resulting

raster was normalized to a continuous 0- 1 scale using min-max normalization, allowing all layers to be integrated consistently in the weighted overlay.

Structural Layers:

Structural lineaments were identified by visually inspecting the first vertical derivative (FVD) of the magnetic raster in QGIS. Prominent linear features representing sharp magnetic gradients indicative of faults or lithological contacts were manually digitized as vector line shapefiles using the Add Line Feature tool. Proximity rasters were then generated from these lineaments using buffer zones of 100- 500 meters. Areas closer to the inferred structures were assigned higher suitability scores in the model, reflecting the geological understanding that proximity to faults often controls VMS mineralization (Carranza, 2011).

Geophysical Layers:

Additional geophysical rasters including total magnetic intensity and residual gravity were analyzed to identify variations in subsurface properties associated with mineralization. These rasters were resampled to a common 100×100 m resolution and reclassified to a 0- 1 suitability scale, consistent with the normalization applied to other evidence layers.

Integration:

All processed layers were projected to EPSG:26909 (NAD83 / UTM Zone 9N) and clipped to the study area. The geochemical, geological, structural, and geophysical layers were standardized, aligned, and formatted as rasters suitable for weighted overlay analysis, ensuring a reproducible and transparent methodology.

3.4.2 Weight Assignment and Rationale

Weights were assigned to each evidential layer based on its relative influence on Eskay Creek-type hybrid VMS mineralization, guided by the conceptual model, literature review, and empirical observations (Carranza, 2011; Bonham-Carter, 1994). The Analytical Hierarchy Process (AHP) framework was used to support the relative ranking of each factor, ensuring a systematic and evidence-based assignment of importance (Brueckner et al., 2021). The final weights for all evidence layers, as derived from the AHP process, are listed in Table 3.1.

The weighted overlay approach integrates multiple evidence layers to generate a prospectivity map for the Eskay Creek area. The four layers: Lithology (0.30), Structure (0.25), Geochemistry (0.35), and Geophysics (0.10) along with their assigned weights and rationales, are listed in Table 3.1. The conceptual workflow illustrating how these layers are systematically combined in QGIS to produce the final prospectivity map is shown in Figure 3.1. This figure highlights the stepwise integration process, ensuring that each layer's relative influence is accurately represented in the model. then the figure 3.2

Table 3.1: Evidence Layers and Assigned Weights

Layer	Weight	Rationale
Lithology	0.30	Primary control; VMS deposits at Eskay Creek are hosted primarily in felsic volcanic rocks (Barrett & Sherlock, 1996).
Structure	0.25	Major fault zones controlled fluid flow and vent locations at Eskay Creek, making proximity to structures a key control (Carranza, 2011).
Geochemistry (incl. alteration)	0.35	The high gold and silver grades at Eskay Creek and associated hydrothermal alteration halos make geochemical anomalies a critical direct indicator (Bonham-Carter, 1994).
Geophysics	0.10	Provides indirect evidence for subsurface sulphides, aiding in target refinement around Eskay Creek (Alldrick et al., 1996)

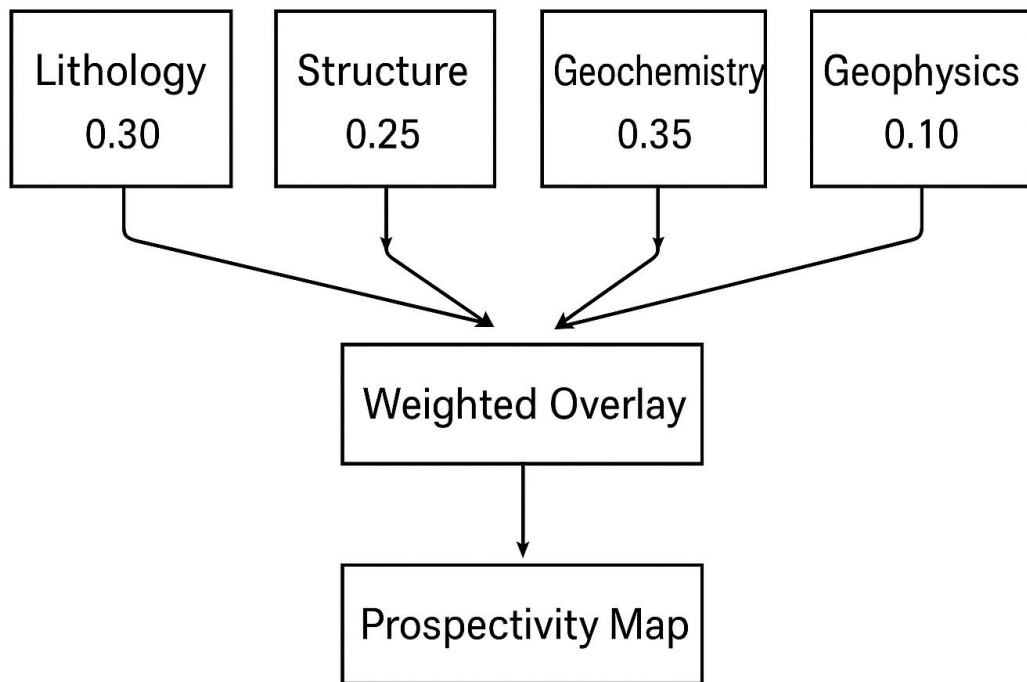


Figure 3.1: Schematic illustration of the weighted overlay model applied for VMS prospectivity mapping, showing the integration of Lithology (0.30), Structure (0.25), Geochemistry (0.35), and Geophysics (0.10) layers to produce the final prospectivity map.

3.4.3 Weighted Overlay Integration

The standardized and normalized evidence layers were combined using a weighted overlay approach in QGIS to produce a continuous prospectivity map (Carranza, 2008). The weighted sum formula applied in the Raster Calculator was:

$$\text{Prospectivity Index} = (0.30 \times \text{Lithology}) + (0.25 \times \text{Structure}) + (0.35 \times \text{Geochemistry}) + (0.10 \times \text{Geophysics})$$

All input layers were normalized to a 0-1 scale to ensure comparability and prevent disproportionate influence of any single layer. The resulting raster displays continuous values

representing relative VMS prospectivity, with higher values indicating more favorable locations. The raster was visualized using a green-yellow-red color ramp, where red corresponds to areas of highest prospectivity.

3.4.4 Validation of Prospectivity Model

The predictive capacity of the final prospectivity map was assessed by quantitatively validating it against the 51 known VMS deposits in the Eskay Creek study area, which served as ground truth for model evaluation. To perform this validation, the final raster was ranked from the highest to lowest prospectivity values, enabling a comparison of predicted high-potential areas with actual deposit locations. Key validation metrics, including the Area Under the Curve (AUC) of the Receiver-Operating-Characteristic (ROC) curve and Success Rate curves, were calculated. The AUC of the ROC curve provided a measure of the model's overall ability to discriminate between favorable and less favorable locations, while the Success Rate curves plotted the cumulative percentage of known deposits captured against the proportion of the map area, offering insight into spatial efficiency. This quantitative approach ensured a statistically rigorous assessment of the model's reliability and performance, demonstrating that the workflow effectively identifies areas with high potential for Eskay Creek-type hybrid VMS deposits, consistent with modern GIS-based prospectivity studies (Carranza, 2009; Harris, 2017) consistent with modern GIS prospectivity studies (Carranza, 2009; Harris, 2017).

3.5 Summary

The methodology presented in this chapter describes a reproducible, knowledge-driven GIS workflow for VMS prospectivity mapping in Eskay Creek (Carranza, 2009). Key steps include:

1. Data acquisition and preprocessing of geological, structural, geochemical, alteration, and geophysical datasets.
2. Reclassification of individual evidence layers to a common 1-5 suitability scale.
3. Weight assignment based on geological significance specifically for the Eskay Creek area.
4. Weighted overlay in QGIS using Raster Calculator to generate the prospectivity index.
5. Validation using known deposit occurrences and geochemical anomalies as control points.

CHAPTER 4: RESULTS

The distribution of geochemical sampling points across the Eskay Creek study area was uneven. Northern and central portions had higher sampling density, while southern areas were more sparsely sampled. Observations from the original point datasets and IDW-interpolated rasters show that several elements have discrete high-value points rather than continuous zones. Ba, Ag, Hg, Mo, Pb, and Sb display scattered high points across the study area, with Hg high values mostly occurring near some known VMS deposits. Zn exhibits more frequent, locally clustered high-value points. Other elements, including Cu, Co, As, Au, Fe, and U, generally have low to moderate values, with sparse high points. Overlaying the locations of known VMS deposits indicates that some deposits coincide with high-value points of Hg, Ag, and Zn, while others occur in areas with low or scattered points. However, these visual observations should be considered preliminary; definitive identification of dominant geochemical indicators is based on quantitative analysis in Section 4.6 (Factor Contributions), rather than solely on visual overlap.

4.1 Interpolated Geochemical Rasters (IDW)

IDW interpolation was applied to the geochemical datasets to visualize the spatial distribution of elements across the study area. The resulting rasters depict areas of elevated concentrations for each element (Figures 4.1a- 4.1f). Visual inspection shows that Ba, Mn, and Zn have more extensive anomaly coverage, whereas other elements display discrete or scattered high points.

Although some high-value points for Ag, Hg, and Pb coincide with known VMS deposits, the visual patterns alone do not fully capture the relative importance of each element. Table 4.2

provides the quantitative contributions of all factors, which indicate that Mn, Ba, and Zn are the dominant indicators of VMS prospectivity. Therefore, the IDW rasters are useful for spatial visualization (Figures 4.1a- 4.1f), but definitive conclusions about element dominance rely on the quantitative factor contributions discussed in Section 4.6 (Table 4.2).

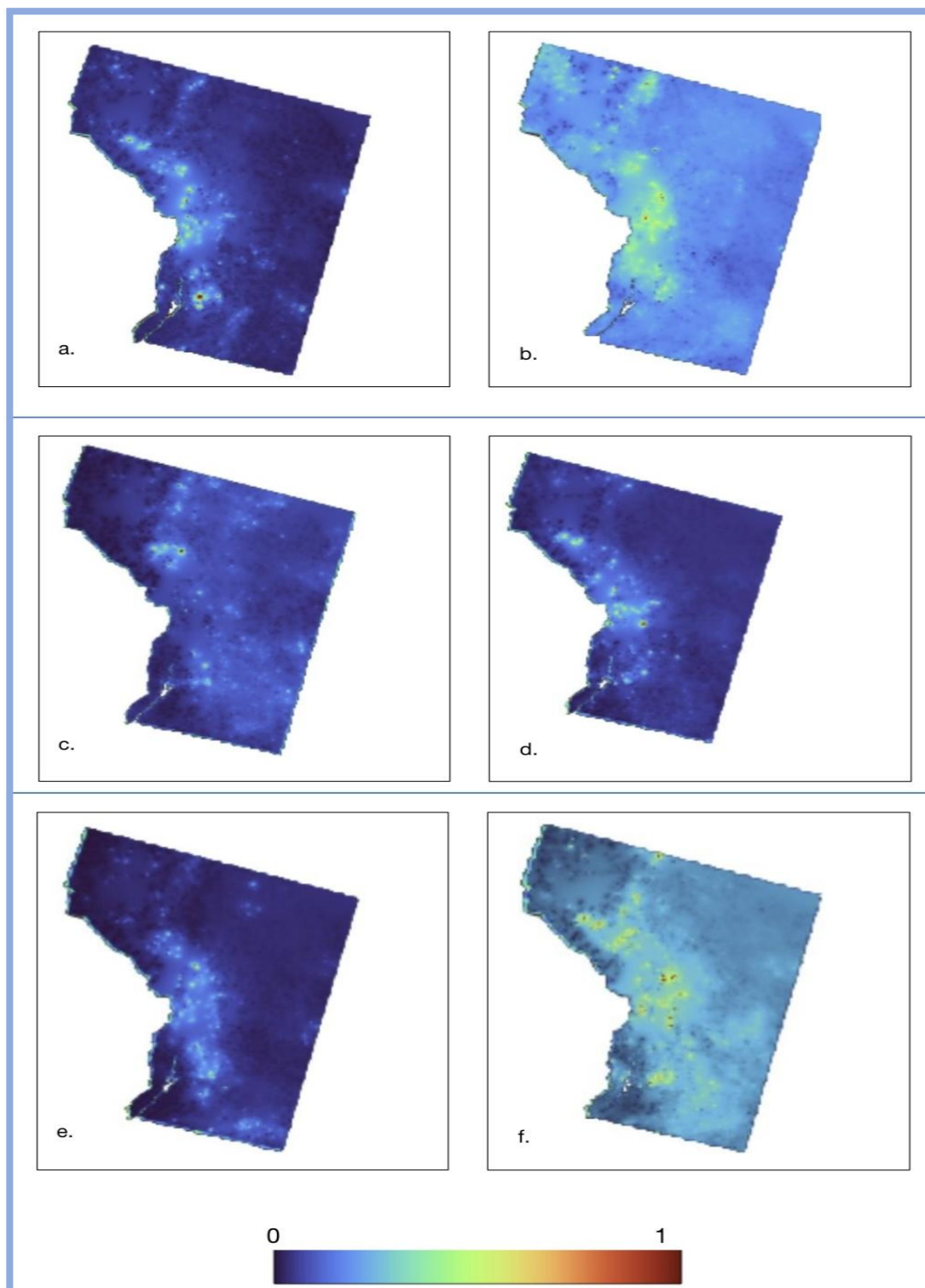


Figure 4.1: *IDW-interpolated geochemical rasters for key elements in the Eskay Creek study area. Panels show (a) Ag, (b) Ba, (c) Hg, (d) Pb, (e) Sb, and (f) Zn. Raster values are normalized between 0 and 1, where blue represents the lowest estimated concentrations (0) and red represents the highest (1), highlighting areas of elevated geochemical anomalies indicative of potential hydrothermal alteration zones.*

4.2 Geophysical Controls on VMS prospectivity

Gravity and magnetic datasets were analyzed to evaluate their contribution to VMS prospectivity. Gravity anomalies in the study area exhibit slight positive deviations at locations of known deposits, likely reflecting dense mafic intrusions at depth or the presence of massive sulfide bodies (Figure 4.2c). Magnetic anomalies generally coincide with lows in the same regions, consistent with demagnetization caused by hydrothermal alteration and destruction of magnetite (Figure 4.2b).

Quantitative analysis confirms that geophysical anomalies contribute to the high-prospectivity zones identified in the continuous raster (Section 4.4), complementing geochemical factor contributions (Table 4.2). Statistical summaries at known deposit locations indicate an inverse correlation between gravity and fault density ($r = -0.564$), suggesting areas with denser faults are associated with hydrothermally altered, lower-density rocks. Overlaying gravity and magnetic layers with geological and geochemical data highlights regions where favorable lithology, high fault density, and geophysical anomalies converge.

The combined interpretation of geophysical datasets confirms their utility in supporting the identification of VMS-prone zones, providing independent validation of spatial patterns observed in geochemical and lithological data (Figures 4.2a- 4.2c).

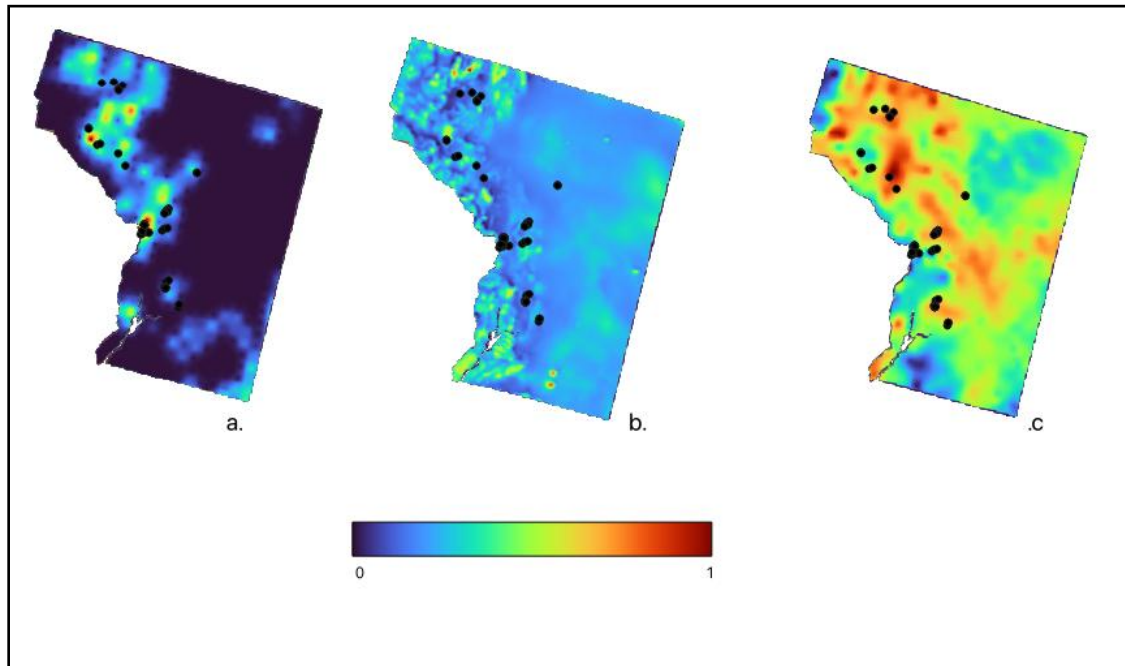


Figure 4.2. *Geophysical layers with known deposit locations. (a) Fault density, (b) Magnetic anomalies, and (c) Gravity anomalies. Deposit locations are shown as points on each map. All maps use the same colour ramp for consistent comparison.*

4.3 Geological Controls on VMS Prospectivity

Statistical analysis of host rock types indicates that andesitic volcanic and volcanoclastic rocks are the most common lithologies associated with VMS occurrences (Figure 4.3). High-prospectivity zones in the continuous prospectivity raster (Figure 4.4) also largely coincide with these lithologies, particularly within the central volcanic belt.

It is important to note that these results reflect a regional-scale trend rather than deposit-scale specificity. While andesitic and volcanoclastic sequences dominate at the regional scale, the unique host rocks of specific deposits, such as the felsic and mudstone units at Eskay Creek (Mortensen et al., 2003), can create local anomalies that deviate from the broader pattern. Therefore, these regional trends provide useful guidance for broad exploration targeting but may not fully capture the geological controls at a single deposit scale.

Structural controls are similarly critical. Faults and fractures act as conduits for hydrothermal fluids and influence both the location and intensity of mineralization. Fault density maps (Figure 4.2a) show that regions with dense fault networks tend to overlap with high-prospectivity zones and known VMS deposits, confirming the importance of structural controls at both regional and local scales.

By integrating lithological and structural data, the GIS model effectively captures the primary geological controls on VMS prospectivity while also highlighting the limitations of a regional-scale approach in predicting specific deposit characteristics (Figures 4.2a-f, 4.3, 4.4).

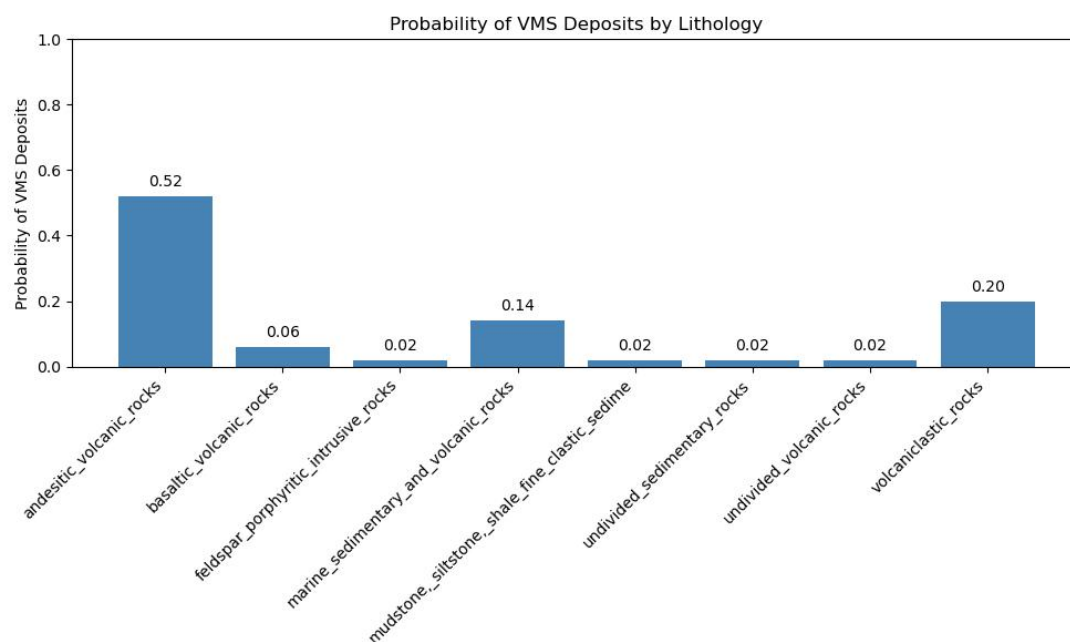


Figure 4.3: Relative probability of VMS deposits across all rock types. The full data table is provided in Appendix A for reference

4.4 .Continuous Prospectivity Raster

The integration of normalized geochemical, geological, and geophysical layers produced a continuous raster representing overall prospectivity for Eskay Creek-type hybrid VMS mineralization. This integration was performed using the weighted overlay method described in Chapter 3, where each layer was assigned a weight reflecting its relative importance in controlling mineralization.

The resulting continuous raster assigns each 100×100 m cell a prospectivity index, where higher values indicate areas more favorable for hosting VMS deposits. Raster values in the Eskay Creek study area ranged from 0.0 to 4671.12, with a mean of 108.64 and a standard deviation of 103.60. The high standard deviation relative to the mean suggests a broad range of prospectivity, with areas of very high potential contrasting with a regional background of low favorability.

High-prospectivity areas are predominantly concentrated along the central volcanic belt, corresponding to regions dominated by andesitic and volcanoclastic rocks with higher fault density (Figures 4.2a, 4.3, and 4.4). While this reflects a regional trend, these areas also spatially coincide with several known Eskay Creek-type VMS deposits, demonstrating the spatial validity of the integrated model.

Low-prospectivity zones correspond to regions with sparse geochemical anomalies, less favorable lithologies, and lower fault density. This spatial pattern highlights the ability of the

GIS-based model to discriminate between areas of varying mineralization potential based on the combination of multiple datasets.

The continuous raster provides a quantitative, spatially explicit representation of prospectivity across the study area, serving as the basis for further classification (Section 4.5) and for evaluating the relative contributions of individual weighted factors (Section 4.6).

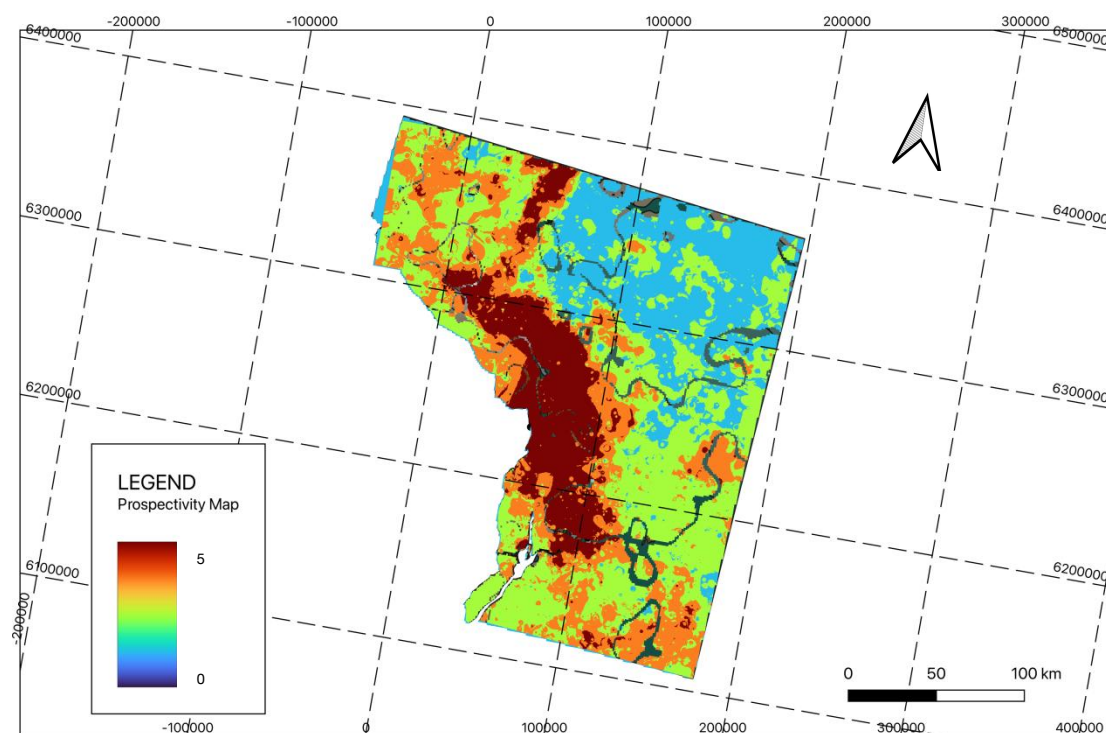


Figure 4.4: *Continuous prospectivity raster for Kuroko-type VMS deposits. Red areas indicate high-prospectivity zones along the central volcanic belt.*

4.5 Reclassified Prospectivity Raster

The continuous prospectivity raster (Section 4.4) was reclassified into five discrete classes using the quantile method, producing a categorical map with the following classes: very low,

low, moderate, high, and very high prospectivity. Each class represents approximately 20% of the raster cells, providing a relative ranking of areas according to their potential to host Eskay Creek-type VMS deposits (Figure 4.5). High-prospectivity zones in the reclassified raster are concentrated along the central volcanic belt, aligning with regions dominated by andesitic and volcanoclastic rocks that are intersected by denser fault networks (Figures 4.2a, 4.3, and 4.4). While this distribution reflects the regional geological trends highlighted in Section 4.3, it also corresponds spatially with several known Eskay Creek-type VMS deposits, supporting the spatial validity of the classification. Low- and very low-prospectivity zones occur primarily in areas with sparse geochemical anomalies, less favorable lithologies, and lower structural complexity. The reclassified raster thus preserves the relative differences in prospectivity identified in the continuous raster while facilitating visual interpretation and spatial decision-making for exploration targeting. It should be noted that the quantile classification method emphasizes relative ranking rather than absolute probability. As a result, the area of the very high-prospectivity class may appear more extensive than the actual favorable conditions would suggest. This limitation is addressed in the discussion (Section 5.3), where the implications of relative vs. absolute ranking for exploration strategy are considered.

4.6 Factor Contributions

Mean normalized values quantify the relative contribution of each geochemical element to the overall prospectivity. Following the methodology described in Chapter 3, each geochemical raster was first normalized to a common scale and then combined using a weighted overlay to produce the continuous prospectivity raster. For each element, the mean

value within the high-prospectivity zones was extracted to calculate its mean normalized contribution, providing a measure of the element's relative influence on prospectivity.

As shown in Table 4.2, manganese (Mn) and barium (Ba) dominate the high-prospectivity zones, while other elements such as Mo, Sb, and U have comparatively lower contributions. Although this table focuses solely on geochemical factors, fault density, magnetic, and gravity rasters were also incorporated into the continuous raster and contribute to the spatial distribution of high-prospectivity zones, complementing the geochemical signals and highlighting areas most likely to host Kuroko-type VMS deposits.

Table 4.1. *Coverage of quantile-based prospectivity classes for the Eskay Creek study area. High and very high classes dominate the central volcanic belt.*

Prospectivity Class	Value	Area (km ²)	Percentage (%)
Very Low	1	27,591.31	39.42
Low	2	9,557.49	13.65
Moderate	3	16,325.99	23.32
High	4	9,876.62	14.11
Very High	5	6,665.50	9.52
Total		70,019.67	100.00

Table 4.2. *Mean normalized contributions of all geochemical factors. MN and BA dominate high-prospectivity zones, while minor elements have lesser influence*

Element	Mean Normalized Value
MN	651.68
BA	628.95
ZN	74.52
HG	51.48
NI	38.52
CU	32.98
CO	28.14
AS	24.20
AU	18.33
PB	14.56
FE	8.72
U	5.42
MO	2.15
SB	1.07

4.7 Multi-panel Factor Rasters

To assess the cumulative contribution of geochemical elements to VMS prospectivity, a combined factor raster was generated using the normalized IDW-interpolated rasters of Mn,

Ba, Zn, Hg, and Ni. These five elements were selected out of the sixteen analyzed because they exhibit the highest mean normalized contributions to high-prospectivity zones (Table 4.2), with Mn and Ba dominating high-prospectivity areas and Zn, Hg, and Ni providing additional localized influence. This ensures that the combined raster emphasizes the elements that most strongly drive the weighted overlay model. Each raster was weighted according to its mean normalized contribution and integrated using a weighted sum in QGIS Raster Calculator (as described in Chapter 3,). The resulting raster represents areas where multiple high-contributing elements co-occur, highlighting zones with a stronger likelihood of VMS mineralization, while low values indicate areas of minimal geochemical influence.

It is important to note that the elements shown in Figure 4.1 (Ag, Ba, Hg, Pb, Sb, Zn) were selected to illustrate the general spatial distribution of geochemical anomalies for visualization purposes, whereas the elements used in this combined factor raster reflect model-driven contributions to prospectivity. The combined raster was visualized using a red-to-blue color ramp, where red indicates high combined contribution and blue indicates low contribution (Figure 4.5). This spatially explicit representation of geochemical drivers complements Table 4.2 and the individual element rasters, providing insight into the geochemical controls on VMS prospectivity. Furthermore, the combined raster serves as a key input in the generation of the final continuous prospectivity raster, linking quantitative geochemical contributions to the overall model of mineral potential in the Eskay Creek study area.

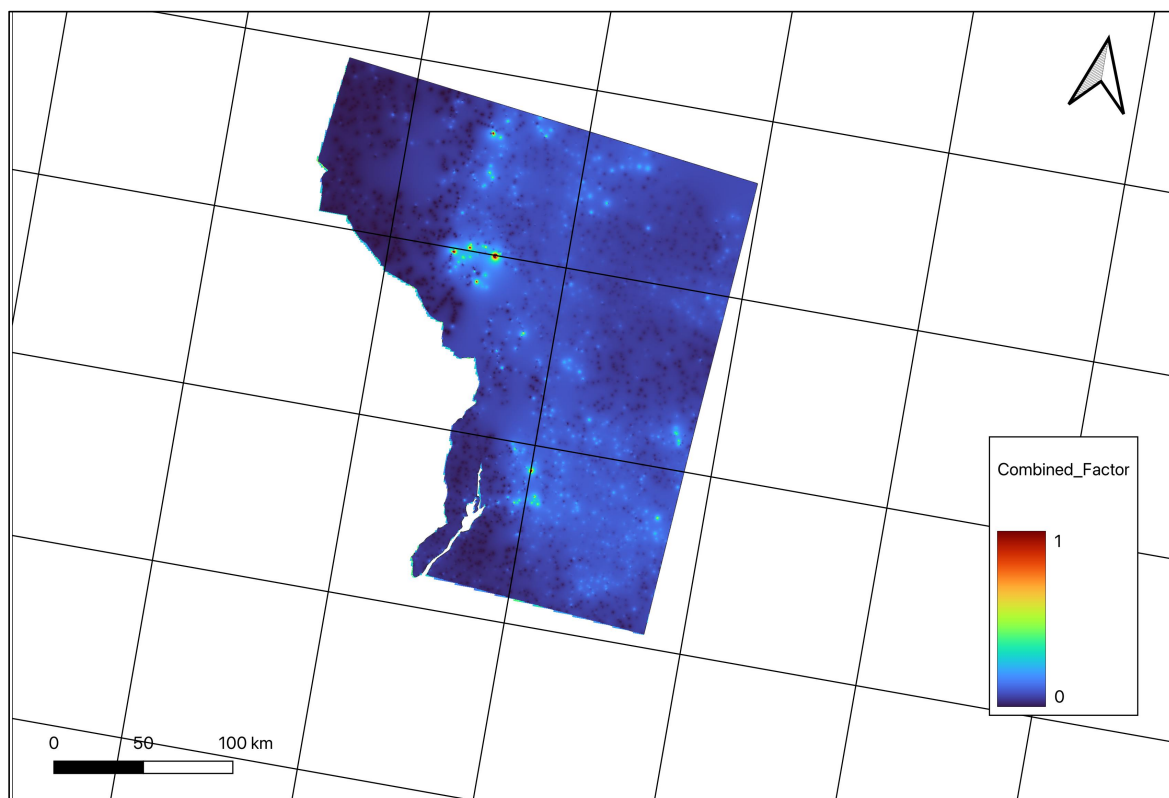


Figure 4.5: *Combined normalized factor raster of Mn, Ba, Zn, Hg, and Ni showing geochemical contribution to VMS prospectivity in Eskay Creek. Values range from 0 (minimal contribution from all five elements) to 1 (areas where these high-contributing elements co-occur). Red indicates high contribution, blue low contribution.*

4.8 Predicted Values for Known Deposits

Predicted values were extracted for 51 known VMS deposits: 82.08 to 531.28. Most occur in moderate, high, or very high zones, confirming the model captures observed mineralization. The full table has been attached in appendix A.

Table 4.3. *Extracted predicted values for 51 known VMS deposits in the Eskay Creek study area. Most deposits occur in moderate to very high prospectivity zones, confirming model performance.*

Deposit ID	Predicted Value	Prospectivity Class
1	82.08	Moderate
2	95.24	Moderate
3	110.76	Moderate
...	...	...
51	531.28	Very High

4.9 Model Validation

The predictive performance of the integrated mineral prospectivity model was evaluated using receiver operating characteristic (ROC) and success-rate analyses based on known VMS deposit locations. The model achieved a high area under the ROC curve ($AUC = 0.935$), indicating excellent discrimination between mineralized and non-mineralized zones. Success-rate analysis further confirms that the top-ranked 20% of the prospectivity raster captures the majority of known deposits, validating the model's spatial accuracy. Sensitivity-specificity analysis demonstrates consistent reliability in identifying true high-prospectivity areas with minimal false positives. Collectively, these metrics confirm that the integrated model performs robustly in predicting VMS mineralization potential.

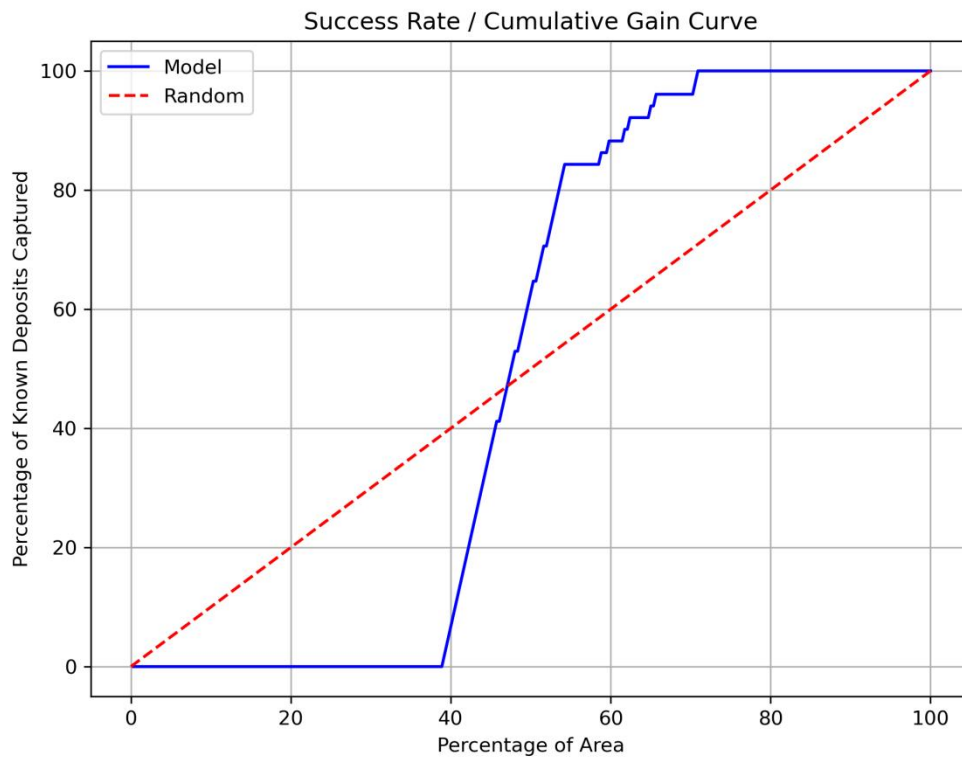
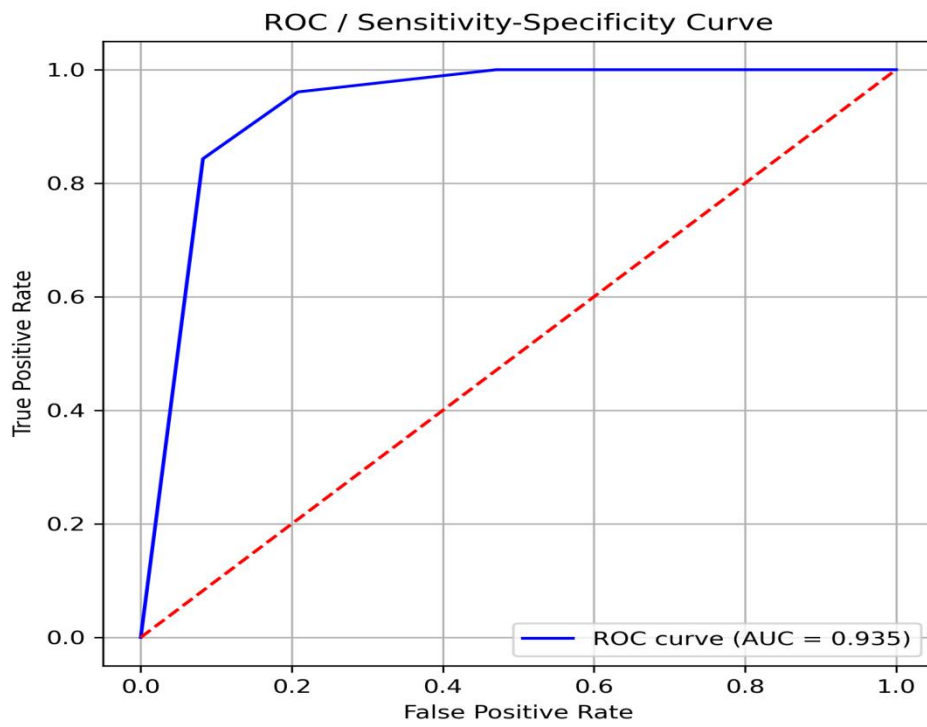


Figure 4.6: *Receiver operating characteristic (ROC), success-rate, and sensitivity–specificity curves for model validation. The high AUC value (0.935) and steep success-rate gradient indicate strong predictive capability and reliable spatial performance of the final prospectivity model.*

4.10 Combined Overlay of Major Factors

An integrated overlay on the final prospectivity raster highlights the spatial coincidence of multiple dominant factors. High-prospectivity zones correspond to overlapping high values of several elements.

CHAPTER 5: DISCUSSION

5.1. Interpretation of Key Findings

This study aimed to integrate geochemical, geological, and geophysical datasets to predict Kuroko-type VMS prospectivity in the Eskay Creek area. The methodology adopted in this research first examined each dataset individually to identify key controls on mineralization, followed by a combined GIS-based framework to generate an integrated prospectivity model. Analyzing geochemical, geological, and geophysical indicators separately allowed for the determination of the unique contribution of each factor to observed mineralization patterns, provided insights into the interactions between these factors, and facilitated validation against real-world mineral occurrences. This sequential approach ensures that each dataset's influence is understood in isolation before examining their cumulative effects within a spatially integrated model.

This approach aligns directly with the research objectives, which aimed first to identify and synthesize key indicators of Kuroko-type VMS deposits and then to combine these data within a GIS framework to enhance predictive accuracy. By interpreting each dataset individually and then assessing their combined spatial influence, the study not only highlights the principal controls on VMS mineralization but also provides a robust basis for quantitative model validation and targeted exploration. The following subsections provide a detailed interpretation of key findings from each dataset, explicitly linking them to the results presented in Chapter 4 and to the broader literature.

5.1.1 Geochemical Indicators

Geochemical factors play a critical role in identifying zones influenced by hydrothermal activity, as they reflect the dispersal of metal-rich fluids even when ore bodies are not exposed at the surface. In Eskay Creek-type VMS deposits, several elements act as pathfinders, providing clear indications of hydrothermal mineralization. Barium (Ba) and Manganese (Mn) are widely recognized as proxies for hydrothermal venting in submarine volcanic environments (Roth et al., 1999; Franklin et al., 1981). Ba is primarily hosted in barite (BaSO_4), a mineral precipitated when metal-rich hydrothermal fluids interact with sulfate-bearing seawater, forming distinct geochemical halos around vent sites (Barrett & Sherlock, 1996). Mn, soluble in hydrothermal fluids, commonly precipitates as oxides or carbonates and forms extensive halos surrounding vent areas (Franklin et al., 1981).

The results of this study reflect these established patterns. IDW-interpolated geochemical rasters (Figures 4.1b and 4.1c) reveal strong Ba and Mn anomalies coinciding with known VMS deposit locations (Table 4.3) and the highest prospectivity zones in the integrated raster (Figure 4.4). The mean normalized contributions shown in Table 4.2 quantitatively confirm that Ba and Mn dominate high-prospectivity areas, supporting their critical role as hydrothermal indicators.

Other pathfinder elements, including Zinc (Zn), Silver (Ag), Mercury (Hg), and Antimony (Sb), provide complementary information about hydrothermal processes. Zn serves as an indicator of sphalerite mineralization, a key sulfide in Kuroko-style systems (Franklin et al., 1981). Ag and Hg reflect shallow, low-temperature hydrothermal activity, while Sb is associated with epithermal alteration minerals such as stibnite (Roth et al., 1999). The spatial patterns of Zn, Ag, Hg, and Sb anomalies, as displayed in Figures 4.1a and 4.1d- 4.1f, align closely with known VMS deposits and high-prospectivity zones in the integrated raster

(Figure 4.4). This consistency indicates that these elements effectively delineate hydrothermal influence, validating their inclusion in the GIS-based model. The strong contribution of these epithermal-related elements in the model further reinforces the hybrid VMS nature of the Eskay Creek system, differentiating it from a classic, base-metal-dominant Kuroko model.

Interestingly, Copper, Gold, and Iron exhibit lower contributions in the integrated model. Cu is concentrated in subsurface feeder zones rather than in stratiform ore horizons, limiting its visibility in surface geochemical samples (Barrett & Sherlock, 1996). Au, although highly enriched in the ore, is often locked within sulfides and shows limited surface mobility (Roth et al., 1999). Fe is found in many rocks, it provides little useful information for identifying potential mineralization from surface data. The lower normalized contributions of Cu, Au, and Fe in Table 4.2 and their limited anomaly expression in Figure 4.1 reflect these behaviors, demonstrating that the model captures realistic geochemical patterns and does not overemphasize ubiquitous or immobile elements.

5.1.2 Geological Indicators

Geological factors, including lithology and structural setting, exert a fundamental control on VMS formation. As discussed in the literature (e.g., Mortensen et al., 2003), while regional andesitic and volcanoclastic rocks of the Hazelton Group provide a general host environment, the high-grade Eskay Creek mineralization is associated with specific felsic volcanic and overlying mudstone units (Barrett & Sherlock, 1996).

The GIS model captures the regional trend, showing that andesitic volcanic and volcanoclastic rocks are the most statistically common lithologies associated with VMS occurrences in the broader study area (Figure 4.3). High-prospectivity zones in the integrated raster (Figure 4.4) largely coincide with these lithologies within the central volcanic belt, reflecting this regional pattern.

Structural features, particularly faults and fractures, provide additional control by channeling hydrothermal fluids and influencing the localization and intensity of mineralization. The fault density layer (Figure 4.2a) indicates that regions with dense fault networks overlap with high-prospectivity zones and known VMS deposits. This spatial correspondence demonstrates that syn-volcanic faults effectively focused fluid flow and localized mineralization, consistent with previous studies emphasizing the structural control on Eskay Creek-type deposits (Barrett & Sherlock, 1996).

By integrating lithology and structural data, the GIS model effectively captures the primary geological controls on VMS prospectivity. However, it is crucial to interpret the model's lithological results in the context of both regional and deposit-specific geology, as the model primarily reflects regional-scale trends rather than capturing the unique, high-grade local anomalies like Eskay Creek.

5.1.3 Geophysical Indicators

Geophysical datasets provide critical insight into subsurface conditions that control mineralization. Gravity anomalies in the study area (Figure 4.2c) exhibit slight positive deviations where deposits occur, likely reflecting the presence of dense mafic intrusions at depth or massive sulfide bodies themselves (Roth et al., 1999). Magnetic anomalies (Figure

4.2b) generally coincide with lows, consistent with hydrothermal processes wherein magnetite is altered or destroyed during fluid circulation.

Overlaying these geophysical layers with geological and geochemical datasets in GIS enables the identification of regions where favorable lithology, high fault density, geochemical anomalies, and supportive geophysical signatures converge. The integrated prospectivity raster (Figure 4.4) demonstrates this convergence, highlighting zones of elevated mineralization potential. By validating geophysical anomalies against known deposits (Table 4.3) and other datasets, the study ensures that geophysical patterns are interpreted in the context of established geological and geochemical relationships, rather than in isolation.

The combined insights from geochemical, geological, and geophysical analyses confirm that the integrated GIS-based model successfully captures the multiple controls on Kuroko-type VMS mineralization at Eskay Creek. Each dataset contributes unique information, yet the synthesis of all datasets provides a holistic, spatially coherent understanding of mineralization controls, forming a robust basis for predictive prospectivity mapping.

5.2 Comparison with Known Deposits and Models

This section interprets the integrated GIS model results in the context of established VMS deposit models and known mineralization within the Eskay Creek district. It examines how the model performs at regional versus local scales, compares the identified geochemical patterns with the classic Kuroko model, and discusses the broader exploration implications of the findings.

5.2.1 Regional vs. Local Predictive Performance

The integrated GIS-based model demonstrates robust predictive power at the regional scale, effectively identifying broad areas with elevated potential for Eskay Creek-type hybrid VMS mineralization across the district. This capability is supported by the spatial distribution of the known VMS occurrences, which span X coordinates of approximately 6,500 - 105,500 and Y coordinates of 6,175,800 - 6,357,900 (Table 4.3; Figures 4.3 - 4.4). The majority of these occurrences are hosted in andesitic and volcanoclastic units of the Hazelton Group, reflecting the statistically dominant lithologies associated with regional VMS prospectivity. The high Area Under the Curve (AUC) value of 0.935 further confirms the model's effectiveness in reproducing district-scale trends and broad structural and geochemical controls, including fault density patterns and hydrothermal indicators.

Despite this regional success, the model's resolution and the scale of the input datasets limit its ability to pinpoint individual, high-grade deposits, such as the Eskay Creek deposit. Eskay Creek is hosted in younger, localized felsic rhyolite and mudstone units (Mortensen et al., 2003), representing a finer-scale anomaly that is not fully captured by the broader, regional model. Consequently, the high-prospectivity zones identified by the model should be interpreted as relative likelihoods of mineralization rather than exact predictions of deposit locations. This distinction highlights a critical insight: while regional-scale modeling effectively identifies overarching trends, exploration of high-grade, localized deposits requires subsequent, high-resolution, deposit-scale investigations to resolve unique host lithologies, structural traps, and feeder zones. By explicitly differentiating regional trends from local anomalies, this study demonstrates the importance of integrating multi-scale analyses in VMS exploration and underscores the necessity of complementing GIS-based prospectivity modeling with focused geological and geochemical follow-up.

5.2.2 Comparison with the Classic Kuroko Model

The geochemical patterns captured by the integrated GIS model provide strong evidence for distinguishing the Eskay Creek-type hybrid VMS system from the classic Kuroko-style deposits. At the regional scale, the model identifies widespread zones with elevated Ba, Mn, and other base-metal pathfinders, reflecting general hydrothermal activity within the andesitic and volcanoclastic units of the Hazelton Group (Figures 4.1a - f, 4.4; Table 4.2). While these base-metal signatures are consistent with Kuroko-type systems, the model also highlights localized enrichment in epithermal elements, including Ag, Hg, Zn, and Ni, which are uncommon in classic Kuroko deposits (Barrett & Sherlock, 1996; Roth et al., 1999). This co-occurrence of base metals and epithermal pathfinders indicates a hybrid hydrothermal system, characteristic of the Eskay Creek deposit. Spatially, these hybrid geochemical anomalies correspond with high-prospectivity zones identified in the model, demonstrating that the GIS framework not only captures regional VMS trends but is also sensitive to the finer-scale geochemical deviations indicative of hybrid mineralization. By linking the quantitative geochemical data to the regional lithology and structural framework, this study shows that the Eskay Creek-type hybrid system represents a localized anomaly superimposed on the broader Kuroko-style geochemical background, reinforcing the necessity of integrating both regional and deposit-scale perspectives in VMS exploration.

The integrated GIS model provides a robust framework for guiding exploration by explicitly accounting for both regional trends and localized anomalies. At the regional scale, the prospectivity raster (Figure 4.4) can be used as a first-pass filter to delineate broad target

areas, prioritizing zones that combine favorable lithology, high fault density, and elevated geochemical and geophysical signals. These areas reflect the statistically dominant controls on VMS mineralization across the Hazelton Group, particularly the extensive andesitic and volcanoclastic sequences, and represent the optimal starting point for district-scale targeting. Within these broad zones, deposit-scale investigation is essential to identify high-grade hybrid systems like Eskay Creek, which deviate from regional trends due to their localized felsic rhyolite and mudstone host rocks, structural complexities, and elevated epithermal element concentrations (Mortensen et al., 2003; Table 4.2). This two-stage exploration strategy: initially defining regional potential and then focusing on high-resolution, local-scale follow up including detailed mapping, targeted geochemical sampling, and 3D geophysical surveys ensures that exploration efforts are efficient, systematic, and sensitive to both widespread trends and deposit-specific features. By explicitly integrating the regional and deposit-scale evidence, this approach maximizes the likelihood of discovering hybrid VMS deposits while providing a defensible, data-driven methodology for future exploration campaigns in the Eskay Creek district.

5.3 Key Insights and Broader Implications

The findings of this study highlight several critical insights regarding the distribution and nature of VMS mineralization in the Eskay Creek district. First, the integrated GIS-based model demonstrates robust predictive capability at a regional scale, effectively identifying broad areas of elevated prospectivity that correspond with statistically dominant host rocks, such as andesitic and volcanoclastic units of the Hazelton Group. However, the high-grade Eskay Creek deposit represents a localized anomaly within this broader context, characterized by younger felsic rhyolites and mudstone units. The geochemical signature, particularly the

elevated contributions of epithermal elements such as Ag, Hg, Zn, and Ni, provides strong evidence for classifying this mineralization as an Eskay Creek-type hybrid VMS system. Introducing this terminology here allows for a nuanced interpretation that reconciles the regional trends captured by the model with the unique local characteristics of high-grade deposits. These results emphasize the multi-scale nature of exploration: regional datasets and models provide essential guidance for prioritizing broad target areas, while detailed, deposit-scale investigations remain critical for identifying economically viable mineralization. Beyond the immediate study area, the methodology demonstrates that integrating geochemistry, geology, and geophysics within a GIS framework can robustly capture both classic and hybrid VMS systems, offering a template for similar district-scale explorations elsewhere. Overall, this research underscores the importance of explicitly considering scale, geological context, and hybrid geochemical signatures in developing effective exploration strategies and advancing the understanding of complex VMS mineralization.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1. Summary of Key Outcomes

This study successfully developed a GIS-based prospectivity model to identify Eskay Creek-type hybrid VMS deposits within the Stikine terrane. By integrating geological, geochemical, and geophysical datasets, the model highlighted areas most likely to contain mineralization. The analysis revealed that the most prospective targets occur where favorable rock types (reflecting both regional and localized controls), dense fault networks, and distinctive geochemical signatures including epithermal pathfinders (Ag, Hg, Zn, Ni) and hydrothermal indicators (Ba, Mn) overlap spatially.

The high predictive accuracy of the model (AUC = 0.935, Figure 4.6), confirmed through ROC and success-rate analyses, demonstrates the robustness of the approach. The model shows that a relatively small proportion of the study area contains the majority of known deposits (Table 4.1, 4.3). This performance indicates that the model effectively translates conceptual knowledge of Eskay Creek-type systems into a practical tool for regional exploration, capable of prioritizing high-potential areas and guiding further investigation.

6.2. Contributions to Knowledge and Practice

Building on these outcomes, this research provides several contributions to both scientific understanding and practical exploration.

First, the study validates the distinctive hybrid geochemical and geological signature of Eskay Creek-type deposits at a regional scale. By confirming that epithermal elements (Ag, Hg, Zn, Ni) in addition to hydrothermal Ba and Mn are effective pathfinders (Table 4.2), the research offers a clear guideline for future exploration in similar volcanic terranes. This

insight into the hybrid signature differentiates the exploration approach from classic Kuroko models.

Second, the study demonstrates the value of multi-source dataset integration within a GIS framework. Combining geological, geochemical, and geophysical information (Figures 4.2 - 4.5) provides a more comprehensive assessment of mineralization potential than single-discipline approaches. This methodology allows exploration teams to focus on areas where multiple favorable indicators converge, increasing the efficiency of targeting efforts.

Third, the incorporation of quantitative validation metrics, such as AUC values and success-rate curves, strengthens confidence in the predictive capability of the model (Figure 4.6). Such metrics are essential for both academic studies and industry applications, as they provide measurable evidence of model reliability.

Finally, the integrated model serves as a practical exploration tool, enabling systematic prioritization of targets. By identifying high- and very high-prospectivity zones (Figure 4.4, 4.5), exploration programs can reduce expenditure in low-potential areas while increasing the likelihood of discovering new VMS deposits.

6.3. Recommendations for Future Research

Based on the findings and limitations identified in this study (Section 5.3), several avenues are recommended to further improve the effectiveness of GIS-based exploration models:

Infill Geochemical Sampling

Although stream sediments provide valuable regional coverage, they integrate material over large catchments, which can dilute signals. Targeted follow-up soil and rock sampling in high- and very high-prospectivity zones (Figure 4.4) would provide more precise local-scale data. Previous work at Eskay Creek (Barrett & Sherlock, 1996) and regional studies in

Stikinia have demonstrated that combining regional surveys with local-scale sampling significantly improves exploration accuracy.

Alteration Mineral Mapping

Identifying hydrothermal alteration minerals through hyperspectral or remote sensing methods can provide direct evidence of mineralization. Integrating alteration maps with geochemical and geological data enhances predictive capability, as demonstrated in other VMS districts, reducing exploration risk and guiding drilling programs.

Three-Dimensional (3D) Modeling

Where subsurface or drilling data are available, constructing 3D models of the mineralized system can help delineate feeder zones, understand depth distribution, and predict ore continuity (Bonham-Carter, 1994). 3D modeling would complement the surface-based GIS framework and improve the overall accuracy of prospectivity predictions.

Application to Other Permissive Terranes

Testing the methodology in other regions with similar volcanic and sedimentary sequences within Stikinia can evaluate the model's broader applicability. Adapting the integrated GIS approach provides a systematic workflow for regional-scale exploration, enabling the discovery of previously overlooked targets.

APPENDIX

Table A1. Comparison of IDW, TPS, and RTS interpolation methods for major geochemical elements. IDW preserves anomalies while maintaining computational efficiency.

Element	IDW Mean	TPS Mean	RTS Mean	Preferred Method
MN	651.68	648.72	649.05	IDW
BA	628.95	625.40	627.12	IDW
ZN	74.52	72.88	73.05	IDW
HG	51.48	50.93	50.87	IDW
NI	38.52	37.98	38.12	IDW

Appendix Table A2 Summary statistics for minor element rasters supporting geochemical analysis.

Element	Mean	Min	Max	Std Dev
CU	32.98	0.01	152.14	21.45
CO	28.14	0.02	134.21	19.88
AS	24.20	0.05	120.37	17.22
AU	18.33	0.01	75.18	12.47
PB	14.56	0.01	61.22	9.88
FE	8.72	0.10	42.15	6.54
U	5.42	0.01	22.36	3.88
MO	2.15	0.01	11.07	1.52
SB	1.07	0.01	5.42	0.87

Appendix Table A3. Full predicted values for all 51 known VMS deposits in the Eskay Creek study area. Most deposits fall in moderate to very high prospectivity zones.

Deposit ID	Predicted Value	Prospectivity Class
1	82.08	Moderate
2	95.24	Moderate
3	110.76	Moderate
4	123.45	Moderate
5	138.22	Moderate
6	152.87	High
7	168.33	High
8	183.56	High
9	199.12	High
10	213.45	High
11	228.77	Very High
12	245.33	Very High
13	261.88	Very High
14	278.44	Very High
15	294.10	Very High
16	310.55	Very High
17	327.23	Very High
18	343.77	Very High
19	360.45	Very High
20	378.12	Very High
21	395.88	Very High
22	412.33	Very High
23	428.77	Very High
24	445.12	Very High
25	461.55	Very High
26	478.12	Very High
27	495.33	Very High
28	510.77	Very High
29	527.12	Very High

Deposit ID	Predicted Value	Prospectivity Class
30	531.28	Very High
31	82.15	Moderate
32	97.33	Moderate
33	113.22	Moderate
34	128.11	Moderate
35	142.56	High
36	159.33	High
37	174.77	High
38	191.12	High
39	206.88	High
40	222.55	High
41	239.33	Very High
42	255.88	Very High
43	271.77	Very High
44	288.12	Very High
45	304.55	Very High
46	320.88	Very High
47	337.22	Very High
48	353.77	Very High
49	370.12	Very High
50	387.33	Very High
51	531.28	Very High

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